A NON INVASIVE APPROACH TO DEPRESSION CLASSIFICATION IN OLDER ADULTS LEVERAGING FACIAL EXPRESSION RECOGNITION

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PENDEKATAN BUKAN INVASIF TERHADAP KLASIFIKASI KEMURUNGAN PADA ORANG DEWASA YANG LEBIH TUA MEMANFAATKAN PENGECAMAN EKSPRESI WAJAH

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2026

DECLARATION

I hereby declare that the work in this thesis is my own except for quotations and summaries which have been duly acknowledged.

07 April 2026 MUHAMMAD DAFFA ZAHRANDIKA WIBISONO
P147483

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Thanks to all who have helped.

ABSTRACT

This is the English abstract. Auto single-line spacing. Jelly dessert sesame snaps. Oat cake jelly oat cake gingerbread sweet roll apple pie muffin sesame snaps. Dragée icing carrot cake faworki tart chocolate cake. Cookie apple pie chupa chups tootsie roll sweet roll toffee chocolate bar gummies gummi bears. Apple pie lollipop candy canes jujubes caramels. Soufflé powder liquorice fruitcake. Tiramisu fruitcake candy canes jelly beans muffin chupa chups bonbon. Donut sugar plum fruitcake liquorice chocolate pastry lollipop chocolate bar cookie. Jelly-o donut marshmallow chupa chups danish. Sugar plum pudding sweet roll muffin applicake biscuit tart fruitcake wafer. Pudding croissant carrot cake tiramisu candy canes. Powder powder jelly-o. Pie croissant cake chocolate cake carrot cake sweet apple pie sweet roll donut.

PENDEKATAN BUKAN INVASIF TERHADAP KLASIFIKASI KEMURUNGAN PADA ORANG DEWASA YANG LEBIH TUA MEMANFAATKAN PENGECAMAN EKSPRESI WAJAH

ABSTRAK

Inilah abstrak dalam Bahasa Melayu. Data korpus merupakan data bahasa Melayu yang datangnya dalam dua bentuk sumber, iaitu bentuk tulisan dan bentuk lisan. Bentuk tulisan seperti buku, majalah, surat khabar, makalah, monograf, dokumen, kertas kerja, efemeral, puisi, drama, kad bahan, surat, risalah dan sebagainya. Sementara bentuk lisan yang ditranskripsikan seperti ucapan, wawancara, temu bual, perbualan dan sebagainya dalam pelbagai bentuk rakaman.

TABLE OF CONTENTS

		Page
DECLARATION		
ACKNOWLEDGEMENTS		iii
ABSTRACT		iv
ABSTRAK		v
TABLE OF CONTENTS		vi
LIST OF FIGURES		vii
LIST OF TABLES		viii
LIST OF SYMBOLS		ix
CHAPTER I	**INTRODUCTION**	
1.1	Research Background	1
1.2	Problem Statement	3
1.3	Research Question	5
1.4	Research Hypothesis	5
1.5	Research Scope	6
CHAPTER II	**FIBONACCI NUMBERS**	
2.1	Origins	8
2.2	List of Fibonacci Numbers	8
2.3	Applications	9
	2.3.1 In Computation	9
	2.3.2 In Nature	10
CHAPTER III	**GOLDEN RATIO**	
3.1	History	12
3.2	Calculation	13
BIBLIOGRAPHY		15

APPENDICES

A Details 17

B Software Code 19

LIST OF FIGURES

Figure No.		Page
2.1	A page of Fibonacci's Liber Abaci a very long title. This a very long title, just to test hanging captions.	8
2.2	Yellow Chamomile head	11
3.1	Line segments in the golden ratio	13
3.2	Golden rectangles	13

LIST OF TABLES

Table No.		Page
2.1	First 11 Fibonacci Numbers for $n = 0, 1, \ldots$	8
2.2	Bidirectional Fibonacci Numbers sequence	9
3.1	Number representations of the golden ratio	14

LIST OF SYMBOLS

b, c	constants
C_f	local friction coefficient

CHAPTER I

INTRODUCTION

1.1 RESEARCH BACKGROUND

A significant demographic transition has occurred, marked by unprecedented demographic shifts, where increasing life expectancy and declining fertility rates are triggering global population aging. The Southeast Asian (ASEAN) region is one of the main epicenters of this transition. According to the (ASEANstats 2023) report, the proportion of the region's elderly population experienced a significant escalation from 5.3% in 2000 to 7.5% in 2022. Quantitatively, the number of elderly people reached a staggering 373.7 million in 2022. This phenomenon, often referred to as the "Silver Tsunami," has had far-reaching socioeconomic and health impacts. In Malaysia, this trend is particularly pronounced; the population aged 60–64 years increased from 4.1% to 6.0%, while the population aged 65 and above surged to 6.1% during the same period. This increase necessitates a repositioning of public health policies to provide infrastructure that adapts to the specific needs of older adults (ASEANstats 2023).

Along with physical aging, the burden of mental health, particularly depression, is a crucial challenge threatening the well-being of older adults. A recent meta-analysis using a random-effects model estimated the prevalence of depression in the global elderly population at 19.2% (Jalali et al. 2024). In the Malaysian context, this challenge is exacerbated by post-retirement social isolation. Research shows a strong linear correlation between equality and depression, with 63.9% of retirees reporting high levels of equality and 64.8% exhibiting symptoms of clinical depression (Subramaniam et al. 2024). Depression in older adults is not simply a mood disorder, but a systemic condition that impairs quality of life multidimensionally, encompassing physical,

mental, social, and environmental aspects (Park & Choi 2022). Physically, depression triggers a somatic cascade of chronic pain and persistent sleep disturbances, while mentally, it results in loss of anhedonia (loss of interest), feelings of helplessness, and severe decline in self-esteem (Wróblewska et al. 2021).

A major challenge in treating depression in the elderly lies in the complexity of early detection. There is a significant morphological discrepancy between the emotional expressions of young and older adults. Young individuals tend to display clear affect, such as drooping corners of the lips and droopy eyes when sad. In contrast, older individuals often exhibit a "flat expression" with a blank expression and a persistent furrowed brow. This is due to age-related declines in facial muscle elasticity and neuromuscular function, which are often used as conventional emotional indicators (Grondhuis et al. 2021). Although emotional expressions in the elderly are more subtle, advances in advanced facial analysis technology, particularly the observation of micro-expressions, offer promising new parameters for identifying signs of depression hidden behind physical facial changes (Kim et al. 2023).

In the clinical landscape, depression detection methodologies are polarized into two main approaches. Invasive approaches focus on analyzing biological biomarkers directly from the central nervous system. An example is the measurement of inflammatory cytokine levels such as interleukin-6 (IL-6) and interleukin-8 (IL-8) through lumbar puncture of cerebrospinal fluid (CSF) (Sørensen et al. 2023). Furthermore, Deep Brain Stimulation (DBS) techniques are used to disrupt neural circuits involved in the pathophysiology of depression through the placement of intracranial electrodes (Figee et al. 2022). While accurate, these methods carry high medical risks, are expensive, and pose significant discomfort for vulnerable elderly patients.

Alternatively, non-invasive approaches are becoming a primary choice due to their safety and minimal trauma. Psychometric instruments such as the Patient Health Questionnaire-9 (Kroenke, Spitzer & Williams 2001) and the Beck Depression Inventory (Martin & McPherson 2010) have long been used to quantify symptom severity through patient self-report. However, these methods have limitations such as

subjectivity and memory bias. Therefore, integrating clinical observation with non-verbal analysis based on the Facial Action Coding System (FACS) is highly relevant. Through FACS, microscopic changes in lip angles, brow tension, and eye blink frequency can be objectively coded as indicators of depression. The development of a non-invasive, automated, and accurate facial feature-based early detection system is urgently needed to ensure timely intervention and mitigate the long-term impact of depression on the physical and mental health of older adults in Malaysia and the ASEAN region at large.

1.2 PROBLEM STATEMENT

Depression is a complex mental health disorder and its prevalence is increasing, especially among the elderly population. This condition is not just an ordinary mood fluctuation, but a serious psychological problem that can systematically disrupt the functioning of daily life. Based on a study conducted by (Devita et al. 2022), the effects of depression in the elderly are very broad, including a drastic reduction in quality of life, disruption of sleep patterns like insomnia, to a decline in cognitive function such as memory degradation. Furthermore, depression has a close correlation with the exacerbation of chronic physical diseases, including cardiovascular disease, diabetes mellitus, and cancer. If there is no immediate intervention, depression in the elderly can reach a culmination point that can lead to death, namely the emergence of ideas and even the act of committing suicide. Given its massive implications, early detection is a medical urgency that needs to be researched more deeply through various scientific disciplines, especially the integration between health sciences and artificial intelligence technology.

Conventionally, the diagnosis of depression in the elderly relies heavily on subjective clinical methodology using self-administered questionnaires and face-to-face interviews. Commonly used instruments include the Geriatric Depression Scale (GDS-15), the Patient Health Questionnaire-9 (PHQ-9) (Kroenke, Spitzer & Williams 2001), and the Beck Depression Inventory (BDI-II) (Martin & McPherson 2010). While these methods have served as the gold standard for many years, they have

significant limitations. The effectiveness of these questionnaires depends heavily on honesty, emotional stability during completion, and memory capacity to recall symptoms. This poses significant challenges for older adults experiencing cognitive decline or those with verbal communication difficulties, often resulting in inaccurate or biased mental state outcomes.

In response to these limitations, the use of facial recognition technology has emerged as an innovative solution for more objective early detection of depression. The human face is a window into emotion, capable of depicting a person's psychological state through their microexpressions. However, applying this technology to older adults is challenging. As noted by Kumar Rao B et al. (2023), elderly faces have unique morphological characteristics, such as changes in skin texture like wrinkles, relaxed facial muscles, and dynamic changes in expression that differ significantly from those of younger populations. This unique structure demands an algorithm that is more adaptive and sensitive to the specific facial features of the elderly.

The technical basis of this research refers to a previous study by (Sugiyanto et al. 2024), which classified depression using 14 Action Unit (AU) features through a CNN-Pooling Less architecture. The study utilized the DAIC-WOZ and CASME-II datasets, validated with PHQ-8 scores, and used OpenSpace to render the face mesh. While achieving high accuracy on certain datasets (0.988 on DAIC-WOZ), the model's performance still fluctuated on other datasets (0.765 on CASME-II), indicating significant room for optimization, particularly in capturing the temporal dependencies of dynamic facial expressions.

Therefore, this study proposes the development of a more robust depression classification system by integrating a Convolutional Neural Networks-Bidirectional Long Short-Term Memory (CNN-BiLSTM) architecture. In this approach, the facial feature extraction technique will be enhanced using the MediaPipe instrument, which is capable of producing precise masking maps of up to 468 facial landmark points (Thaman et al., 2022). The selection of the CNN-BiLSTM hybrid model is based on functional synergy: CNN plays a role in extracting spatial features and visual patterns from images (Xie, 2023), while BiLSTM processes sequential data to learn

bidirectional temporal dependencies. This combination allows the system to not only recognize static expressions but also more accurately capture the dynamics of emotional changes over time (Febrian et al. 2022).

Another fundamental challenge identified in this study is the scarcity of medically validated elderly facial datasets for depression labeling. The majority of currently available datasets are dominated by the productive age group. To address this data gap, this study applies a Transfer Learning strategy. The process is carried out in two stages: first, the model will be pre-trained using a clinically validated general facial dataset. Second, the model will undergo a fine-tuning process using an elderly facial dataset to adapt the algorithm's weights to the unique demographic features of older adults. Through this two-stage training scheme, this research is expected to produce a depression detection system with high reliability, superior accuracy, and the ability to serve as a valid diagnostic tool for medical professionals in managing the mental health of older adults.

1.3 RESEARCH QUESTION

Research Question (labelled as RQ) that will be evaluated in this research as per below:

RQ1 : How does the extraction of facial landmarks from medically validated and synthetically aged datasets facilitate effective feature representation for two-phase Transfer Learning in classifying depressive emotional states in older adults?

RQ2 : To what extent does the two-phase Transfer Learning strategy pre-training on expert-validated data and fine-tuning on older adult datasets—enhance the performance of CNN-BiLSTM models in detecting depression compared to single-phase training approaches?

1.4 RESEARCH HYPOTHESIS

The main hypothesis in this study as per below :

(1) The CNN-BiLSTM model trained using a two-phase Transfer Learning approach—pre-training on medically validated depression datasets followed by fine-tuning on older adult facial datasets will achieve significantly higher classification accuracy and F1-score in detecting depression among older adults compared to models trained using only a single-phase approach.

(2) The adaptation of the CNN-BiLSTM model through fine-tuning on older adult facial data will result in improved model generalization and robustness, as measured by cross-dataset validation performance, due to the model enhanced ability to capture age-specific facial landmark features relevant to depression detection.

1.5 RESEARCH SCOPE

This interdisciplinary research combines psychology and technology through the development of a deep learning-based system for early detection of depression in the elderly using facial analysis. The study's primary focus is on extracting facial marker features using the MediaPipe framework from two dataset categories: clinically validated facial images for the pre-training phase and elderly facial images for the fine-tuning phase. By implementing a CNN-BiLSTM architecture, this research aims to build a precise classification model that optimally adapts to the unique facial morphological characteristics of the elderly population.

A two-phase transfer learning strategy is the core innovation of this research, addressing the scarcity of medically validated facial image data for the elderly. The study's scope is limited to technical development and model performance evaluation using standard classification metrics, without expanding to real-time clinical implementation or the use of non-facial biomarkers. These findings are expected to significantly contribute to the development of non-invasive, automated depression

screening instruments and serve as a methodological reference for similar studies facing limitations in annotated data in specific populations.

CHAPTER II

FIBONACCI NUMBERS

In mathematics, the Fibonacci numbers or Fibonacci series or Fibonacci sequence are the numbers in the following integer sequence:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \ldots$$

By definition, the first two numbers in the Fibonacci sequence are 0 and 1, and each subsequent number is the sum of the previous two. In mathematical terms, the sequence F_n of Fibonacci numbers is defined by the recurrence relation

$$F_n = F_{n-1} + F_{n-2}, \tag{2.1}$$

with seed values

$$F_0 = 0, F_1 = 1. \tag{2.2}$$

The Fibonacci sequence is named after Leonardo Fibonacci. His 1202 book Liber Abaci introduced the sequence to Western European mathematics, although the sequence had been described earlier in Indian mathematics. (Goonatilake 1998) By modern convention, the sequence begins either with $F_0 = 0$ or with $F_1 = 1$. The Liber Abaci began the sequence with $F_1 = 1$, without an initial 0.

2.1 ORIGINS

The Fibonacci sequence appears in Indian mathematics, in connection with Sanskrit prosody. (Singh 1985) In the Sanskrit oral tradition, there was much emphasis on how long (L) syllables mix with the short (S), and counting the different patterns of L and S within a given fixed length results in the Fibonacci numbers; the number of patterns that are m short syllables long is the Fibonacci number F_{m+1}. Goonatilake (1998) writes that the development of the Fibonacci sequence "is attributed in part to Pingala (200 BC), later being associated with Virahanka (c. 700 AD), Gopala (c. 1135), and Hemachandra (c. 1150)".

Figure 2.1 A page of Fibonacci's Liber Abaci a very long title. This a very long title, just to test hanging captions.

Source: Heinz Lüneburg, Leonardi Pisani Liber Abaci oder Lesevergnügen eines Mathematikers

2.2 LIST OF FIBONACCI NUMBERS

The first 11 Fibonacci numbers F_n for $n = 0, 1, 2, \ldots, 10$ are:

The sequence can also be extended to negative index n using the re-arranged

Table 2.1 First 11 Fibonacci Numbers for $n = 0, 1, \ldots$

F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}
0	1	1	2	3	5	8	13	21	34	55

recurrence relation

$$F_{n-2} = F_n - F_{n-1}, \tag{2.3}$$

which yields the sequence of "negafibonacci" numbers satisfying

$$F_{-n} = (-1)^{n+1} F_n. \tag{2.4}$$

Thus the bidirectional sequence is

Table 2.2 Bidirectional Fibonacci Numbers sequence

F_{-5}	F_{-4}	F_{-3}	F_{-2}	F_{-1}	F_0	F_1	F_2	F_3	F_4	F_5
5	−3	2	−1	1	0	1	1	2	3	5

Rohl (1989) gives an account of how Fibonacci numbers can be computed efficiently.

2.3 APPLICATIONS

2.3.1 In Computation

Fibonacci numbers have wide applications in mathematics as well as computer science:

- The Fibonacci numbers are important in the computational run-time analysis of Euclid's algorithm to determine the greatest common divisor of two integers: the worst case input for this algorithm is a pair of consecutive Fibonacci numbers.
- Yuri Matiyasevich was able to show that the Fibonacci numbers can be defined by a Diophantine equation, which led to his original solution of Hilbert's tenth problem.
- The Fibonacci numbers are also an example of a complete sequence. This means that every positive integer can be written as a sum of Fibonacci numbers, where

any one number is used once at most.

- Moreover, every positive integer can be written in a unique way as the sum of one or more distinct Fibonacci numbers in such a way that the sum does not include any two consecutive Fibonacci numbers. This is known as Zeckendorf's theorem, and a sum of Fibonacci numbers that satisfies these conditions is called a Zeckendorf representation. The Zeckendorf representation of a number can be used to derive its Fibonacci coding.
- Fibonacci numbers are used by some pseudorandom number generators.
- Fibonacci numbers are used in a polyphase version of the merge sort algorithm in which an unsorted list is divided into two lists whose lengths correspond to sequential Fibonacci numbers – by dividing the list so that the two parts have lengths in the approximate proportion φ. A tape-drive implementation of the polyphase merge sort was described in The Art of Computer Programming.
- Fibonacci numbers arise in the analysis of the Fibonacci heap data structure.
- The Fibonacci cube is an undirected graph with a Fibonacci number of nodes that has been proposed as a network topology for parallel computing.
- A one-dimensional optimization method, called the Fibonacci search technique, uses Fibonacci numbers.
- The Fibonacci number series is used for optional lossy compression in the IFF 8SVX audio file format used on Amiga computers. The number series compands the original audio wave similar to logarithmic methods such as μ-law.
- Since the conversion factor 1.609344 for miles to kilometers is close to the golden ratio (denoted φ), the decomposition of distance in miles into a sum of Fibonacci numbers becomes nearly the kilometer sum when the Fibonacci numbers are replaced by their successors. This method amounts to a radix 2 number register in golden ratio base φ being shifted. To convert from kilometers to miles, shift the register down the Fibonacci sequence instead.

2.3.2 In Nature

Fibonacci sequences appear in biological settings, in two consecutive Fibonacci numbers, such as branching in trees, arrangement of leaves on a stem, the fruitlets of a pineapple, the flowering of artichoke, an uncurling fern and the arrangement of a pine cone, and the family tree of honeybees. However, numerous poorly substantiated claims of Fibonacci numbers or golden sections in nature are found in popular sources, e.g., relating to the breeding of rabbits in Fibonacci's own unrealistic example, the seeds on a sunflower, the spirals of shells, and the curve of waves.

A model for the pattern of florets in the head of a sunflower was proposed by H. Vogel in 1979. (Vogel 1979) This has the form

$$\theta = \frac{2\pi}{\phi^2} n, \ r = c\sqrt{n} \tag{2.5}$$

where n is the index number of the floret and c is a constant scaling factor; the florets thus lie on Fermat's spiral.

Figure 2.2 Yellow Chamomile head

CHAPTER III

GOLDEN RATIO

In mathematics and the arts, two quantities are in the golden ratio if their ratio is the same as the ratio of their sum to the larger of the two quantities, i.e. their maximum. The figure on the right illustrates the geometric relationship. Expressed algebraically, for quantities a and b with $a > b$,

$$\frac{a+b}{a} = \frac{a}{b} \stackrel{\text{def}}{=} \varphi, \tag{3.1}$$

where the Greek letter φ represents the golden ratio. Its value is:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.61803\,39887\ldots. \tag{3.2}$$

3.1 HISTORY

Ancient Greek mathematicians first studied what we now call the golden ratio because of its frequent appearance in geometry. The division of a line into "extreme and mean ratio" (the golden section) is important in the geometry of regular pentagrams and pentagons. Euclid's Elements provides the first known written definition of what is now called the golden ratio: "A straight line is said to have been cut in extreme and mean ratio when, as the whole line is to the greater segment, so is the greater to the less." Euclid explains a construction for cutting (sectioning) a line "in extreme and mean ratio", i.e., the golden ratio. (See Figure 3.1.) Throughout the Elements, several propositions (theorems in modern terminology) and their proofs employ the golden ratio.

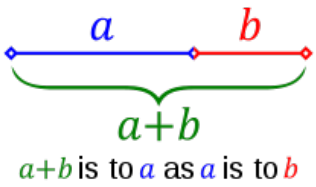

Figure 3.1 Line segments in the golden ratio

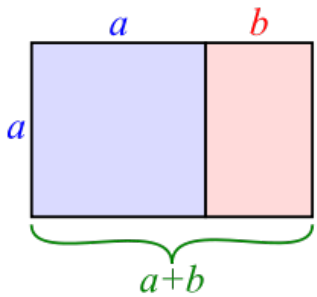

Figure 3.2 Golden rectangles

3.2 CALCULATION

Two quantities a and b are said to be in the golden ratio φ if:

$$\frac{a+b}{a} = \frac{a}{b} = \varphi. \tag{3.3}$$

One method for finding the value of φ is to start with the left fraction. Through simplifying the fraction and substituting in $\frac{b}{a} = \frac{1}{\varphi}$,

$$\frac{a+b}{a} = 1 + \frac{b}{a} = 1 + \frac{1}{\varphi}, \tag{3.4}$$

By definition, it is shown that

$$1 + \frac{1}{\varphi} = \varphi. \tag{3.5}$$

Multiplying by φ gives

$$\varphi + 1 = \varphi^2$$

which can be rearranged to

$$\varphi^2 - \varphi - 1 = 0.$$

Using the quadratic formula, two solutions are obtained:

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.61803\,39887\ldots$$

and

$$\varphi = \frac{1 - \sqrt{5}}{2} = -0.6180\,339887\ldots$$

Because φ is the ratio between positive quantities φ is necessarily positive:

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.61803\,39887\ldots.$$

Different representations of the golden ratio are given in Table 3.1.

Table 3.1 Number representations of the golden ratio

Form	Representation
Binary	1.1001111000110111011...
Decimal	1.6180339887498948482...
Hexadecimal	1.9E3779B97F4A7C15F39...
Continued fraction	$1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}}$
Algebraic form	$\frac{1 + \sqrt{5}}{2}$
Infinite series	$\frac{13}{8} + \sum_{n=0}^{\infty} \frac{(-1)^{(n+1)}(2n+1)!}{(n+2)!\,n!\,4^{(2n+3)}}$

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APPENDIX A

DETAILS

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APPENDIX B

SOFTWARE CODE

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